Apparatus to Paul Gordan, *De linea geodetica* (1862)

Integrated candidate: physical units 1–47

This restrained apparatus is separate from the standalone Latin and English editions. It records one explicit disposition for every physical unit, copy-matter exclusions, and source-critical readings established by direct authority-pixel replay. The diplomatic Latin preserves printed anomalies; the faithful English retains their mathematical content without silently substituting conjectural corrections.

Unit dispositions

PU	Printed	Role	Disposition	Note
1	–	extrinsic wrapper	EXCLUDED; provenance/topology only	1
2	unnumbered	title page	ACCEPTED intrinsic text; copy marks excluded	2
3	–	blank witness	BLANK position retained; copy marks excluded	3
4	unnumbered	dedication	ACCEPTED	4
5	unnumbered	dedication continuation	ACCEPTED	4
6	1	printed page 1	ACCEPTED; source-verified mathematical body	–
7	2	printed page 2	ACCEPTED; source-verified mathematical body	–
8	3	printed page 3	ACCEPTED; source-verified mathematical body	–
9	4	printed page 4	ACCEPTED; source-verified mathematical body	–
10	5	printed page 5	ACCEPTED; source-verified mathematical body	5
11	6	printed page 6	ACCEPTED; source-verified mathematical body	–
12	7	printed page 7	ACCEPTED; source-verified mathematical body	–
13	8	printed page 8	ACCEPTED; source-verified mathematical body	–
14	9	printed page 9	ACCEPTED; source-verified mathematical body	–
15	10	printed page 10	ACCEPTED; source-verified mathematical body	6
16	11	printed page 11	ACCEPTED; source-verified mathematical body	7
17	12	printed page 12	ACCEPTED; source-verified mathematical body	8
18	13	printed page 13	ACCEPTED; source-verified mathematical body	–
19	14	printed page 14	ACCEPTED; source-verified mathematical body	–
20	15	printed page 15	ACCEPTED; source-verified mathematical body	–
21	16	printed page 16	ACCEPTED; source-verified mathematical body	–
22	17	printed page 17	ACCEPTED; source-verified mathematical body	–
23	18	printed page 18	ACCEPTED; source-verified mathematical body	9
24	19	printed page 19	ACCEPTED; source-verified mathematical body	10
25	20	printed page 20	ACCEPTED; source-verified mathematical body	11
26	21	printed page 21	ACCEPTED; source-verified mathematical body	–
27	22	printed page 22	ACCEPTED; source-verified mathematical body	–
28	23	printed page 23	ACCEPTED; source-verified mathematical body	–
29	24	printed page 24	ACCEPTED; source-verified mathematical body	–
30	25	printed page 25	ACCEPTED; source-verified mathematical body	–
31	26	printed page 26	ACCEPTED; source-verified mathematical body	–
32	27	printed page 27	ACCEPTED; source-verified mathematical body	–
33	28	printed page 28	ACCEPTED; source-verified mathematical body	12
34	29	printed page 29	ACCEPTED; source-verified mathematical body	–
35	30	printed page 30	ACCEPTED; source-verified mathematical body	–
36	31	printed page 31	ACCEPTED; source-verified mathematical body	13
37	32	printed page 32	ACCEPTED; source-verified mathematical body	14
38	33	printed page 33	ACCEPTED; source-verified mathematical body	–
39	34	printed page 34	ACCEPTED; source-verified mathematical body	–
40	35	printed page 35	ACCEPTED; source-verified mathematical body	–
41	36	printed page 36	ACCEPTED; source-verified mathematical body	15
42	37	printed page 37	ACCEPTED; source-verified mathematical body	–
43	38	printed page 38	ACCEPTED; source-verified mathematical body	16
44	39	printed page 39	ACCEPTED; source-verified mathematical body	17
45	40	printed page 40	ACCEPTED; source-verified mathematical body	18
46	–	Vita	ACCEPTED intrinsic biography; copy footer excluded	19
47	–	Theses	ACCEPTED intrinsic terminal theses; copy footer excluded	20

Numbered notes

1. **PU001 (provider wrapper).** The complete Google Books reproduction notice, logo, QR code, URL, and rule are extrinsic copy/digitizer matter. They remain in the authority evidence and are excluded from both scholarly editions.

2. **PU002 (title).** The handwritten accession or shelfmark and the "Digitized by Google" footer are copy-specific and excluded. The dissertation title, defence information, opponents, place, and printer are intrinsic and retained. The title rules and central ornament are conservatively reconstructed as editable typographic objects.

3. **PU003 (blank witness).** No authorial text is visible. The blank physical position is retained as a blank page; speckle, edge marks, and the digitizer footer are not reproduced.

4. **PU004–PU005 (dedication).** The Latin abbreviation *D. D. D.* is retained. The English layer expands its conventional force as "he gives, dedicates, and devotes" rather than treating the expansion as Latin source text.

5. **PU010 / printed p.5.** The first elliptic-coordinate variational display visibly reads $1 - e^2 \cos l$, without a superscript on cos. Since nearby formulas use $1 - e^2 \cos^2 l$, the missing exponent is probably a printing error. The diplomatic Latin and faithful English preserve the printed reading; the probable correction is recorded only here.

6. **PU015 / printed p.10.** In the inverse series for x, the authority clearly reads $-\frac{33}{32}p^2$. The inherited p^3 reading was rejected as a transcription error, and the source reading is restored in both editions.

7. **PU016 / printed p.11.** The first-variation symbol carries subscript 1 and the second variation carries subscript 2. The inherited omission of the first subscript was rejected. The printed denominator exponent is retained in its mixed form $1\frac{1}{2}$ rather than silently regularized to $\frac{3}{2}$.

8. **PU017 / printed p.12.** The transformed second-variation denominator again prints the mixed exponent $1\frac{1}{2}$. The source apostrophe in *integralis'* is retained diplomatically.

9. **PU023 / printed p.18.** The dense U-reduction was checked directly against the 400-dpi authority replay. In particular, after $u_1 = 0$ the derivative brace is ordered

$$\frac{u_2}{v_2}\left(\frac{\sin\varphi}{v}\right)'_2 - \frac{\sin\varphi_2}{v_2}\left(\frac{u}{v}\right)'_2 .$$

   The accepted editable formula preserves that ordering.

10. **PU024 / printed p.19.** The theorem visibly prints $\psi < \psi'$ in its first condition but $\psi > \psi$ in the second. The latter is almost certainly a dropped prime, but it is preserved in the diplomatic and English layers. The conjectural $\psi > \psi'$ correction is confined to this note.

11. **PU025 / printed p.20.** The source uses an uppercase Greek Π in the interval "between 0 and 2Π," while the surrounding formulas predominantly use π. The uppercase form is preserved in the diplomatic layer.

12. **PU033 / printed p.28.** Fresh 400-dpi replay establishes the first E-series line as $8q + 16q^2 + 32q^3 \ldots$. The inherited omission of the exponent in $16q^2$ was rejected in both editions.

13. **PU036 / printed p.31.** The extension formula visibly has the coefficient $4q/(1-2q)^3$. A later inherited variant with square rather than cube was rejected.

14. **PU037 / printed p.32.** In $\cos A \operatorname{tg} o = \operatorname{tg} x$ and in the immediately following right-hand $o + 4iq^2 \cdots$, the character is a small Latin o, not Greek σ and not the numeral zero. Both visible occurrences are retained in each edition.

15. **PU041 / printed p.36.** The 3×3 cosine determinant and the three subsequent determinant reductions are reconstructed as editable mathematics. No diagram or image fallback is used.

16. **PU043 / printed p.38.** The text introduces the right triangle BCD and the relation $BD/CD = \operatorname{tg} BCD$ verbally and algebraically; the authority page contains no drawn triangle. No figure is missing from either edition.

17. **PU044–PU045 / printed pp.39–40.** The source prints “e formula:” once and then uses ditto marks in the left column. The Latin preserves those source-visible marks typographically. The English translates the first phrase as “from the formula:” and preserves the subsequent ditto convention.

18. **PU045 / printed p.40.** The numerical example visibly prints $x_1 = 149°38′429{,}65″$. The anomalous seconds value is preserved diplomatically and in the faithful English layer; no normalization to minutes and seconds is made.

19. **PU046 (Vita).** Direct replay restores the printed forms *terras exsternas* and the name *Frankenhein*. The English translates the former as “foreign lands” and retains the latter name. The Google digitizer footer is copy-specific and excluded; the terminal ornament is reconstructed editably.

20. **PU047 (Theses).** All four theses, the title’s terminal dot, and the closing ornament are intrinsic and retained. Speckle, edge marks, and the “Digitized by Google” footer are copy-specific and excluded.